

Comparative Co-Presence Network Analysis of the New Testament Gospels and the Qur'an

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Abstract

This study seeks to analyze textual data from the four New Testament Gospels (Matthew, Mark, Luke, and John) and the Surahs of the Quran to construct and compare co-presence networks. The goal of this project is to evaluate how the distinct literary styles of chronological narratives and thematic discourses alter the structure of the resulting networks. Because the texts differ significantly in length and structure, the methodology requires data normalization, clustering coefficient comparisons, and degree distribution analysis to quantify structural similarities and differences.

1 Introduction

Practitioners of Abrahamic religions make up over half the world's population ¹, and the holy texts of these faiths serve as the cornerstone for their respective beliefs and cultural traditions. Despite historical and contemporary contention between Christianity and Islam, the New Testament Gospels and the Qur'an share a deeply interconnected lineage stemming from Abraham. Shared figures bridge them conceptually, but the texts themselves are structured in fundamentally different ways. The four New Testament Gospels (Matthew, Mark, Luke, and John) are framed primarily as chronological narratives, whereas the Surahs of the Qur'an are formatted as a largely thematic discourse.

This study analyzes textual data from both the Bible and Qur'an to construct, visualize, and compare their underlying character networks. Specifically, this study evaluates how these distinct literary styles alter the resulting graph topology among shared entities. Because the texts differ significantly in length, scope, and structural organization, the methodology incorporates data normalization, clustering coefficient comparisons, and degree distribution analysis to quantify structural similarities and differences. Data for this study is drawn from the BibleData repository for the Gospels [1] and the Sahih International English translation via the Qur'an API [2].

This paper seeks to address three central research questions:

- Does a thematic text structure exhibit different network characteristics compared to a chronological narrative?
- How do the centrality metrics of shared characters shift between the two texts?
- Do both networks exhibit power-law degree distributions despite their fundamental structural differences?

Ultimately, this comparative network analysis serves to show that the shared principal figures rise above the structural settings of their respective texts. Beyond religious and literary divides, they represent a shared foundation of human hope and eternal faith.

2 Literature Review

In attempting to analyze textual data, network analytics provides an empirical perspective to the task. Rather than characters being classified as "protagonists", their betweenness centrality can be calculated and visualized.

Social networks imply an interaction between two characters. A co-presence network emphasizes the relationship between characters rather than requiring explicit conversations. It leverages the literary style of a chronological

¹According to the *World Population Review*, 32% of the global population identify as Christian, 25% Muslim and 0.2% Jewish

narrative. Implicit relationships are also important.

Religious texts written in ancient languages present a problem for modern natural language processing tools. It's hard to identify who a "PERSON" might be when they're given multiple names in the span of the same text. Additionally, determining a threshold for what constitutes an implicit relationship between characters is challenging [3]. This also introduces the deliniation between social, co-presence, and co-occurrence networks.

Topics in social network analytics: degree or valency, centrality metrics, clustering coefficients, diameter, degree, density, and ego networks [4]. These metrics transform qualitative relationships between characters into mathematical terms ripe for comparison.

SNA done in the context of Luke's gospel [5].

Analyzed all four gospels, unsurprisingly Jesus is the hub for all four [6]

3 Methodology

Utilizing data from [1] and [2]

Explain translation vs. transliteration: used english Sahih International translation

How to define textual proximity for edges? - the bibledata already defines relationships within verses - Qur'an data preprocessing was set up to mirror that, co-presence established within each verse

3.1 Global Topology

Nodes Edges Density Average Clustering Coefficient Average Path Length Diameter

3.2 Centrality

Degree Centrality Degree is a simple centrality measure that counts how many neighbors a node has. A node is important if it has many neighbors, or, in the directed case, if there are many other nodes that link to it, or if it links to many other nodes.

$$C_D(v) = \frac{\deg(v)}{N-1}$$

- $\deg(v)$: Number of edges connected to node v
- N : Total number of nodes in the network

Closeness Centrality In a connected graph, the normalized closeness centrality (or closeness) of a node is the average length of the shortest path between the node and all other nodes in the graph. Thus the more central a node is, the closer it is to all other nodes.

$$C'_B(v) = \frac{2}{(N-1)(N-2)} \sum_{s \neq v \neq t} \frac{\sigma_{st}(v)}{\sigma_{st}}$$

- σ_{st} : Total number of shortest paths from node s to node t
- $\sigma_{st}(v)$: Number of those shortest paths that pass through node v

Betweenness Centrality Betweenness centrality quantifies the number of times a node acts as a bridge along the shortest path between two other nodes.

$$C_C(v) = \frac{N-1}{\sum_{u \neq v} d(v, u)}$$

- $d(v, u)$: Shortest path distance (geodesic distance) between node v and node u

Eigenvector Centrality Eigen-centrality is a prestige score which measures the node's influence. It is a nodal attribute, as is the case with all other centrality measures. Connections to high-scoring nodes contribute more to the score of the node in question than equal connections to low-scoring nodes. A high Eigen-centrality means the node is connected to many nodes with high scores.

$$\lambda x_v = \sum_{u \in N(v)} x_u = \sum_{u=1}^N A_{vu} x_u$$

3.3 Power-Law Distribution

A power law is a functional relationship between two quantities, where a relative change in one quantity results in a relative change in the other quantity proportional to a power of the change, independent of the initial size of those quantities

$$f(x) = \alpha x^{-k}$$

- In real networks, properties are often power-law distributed

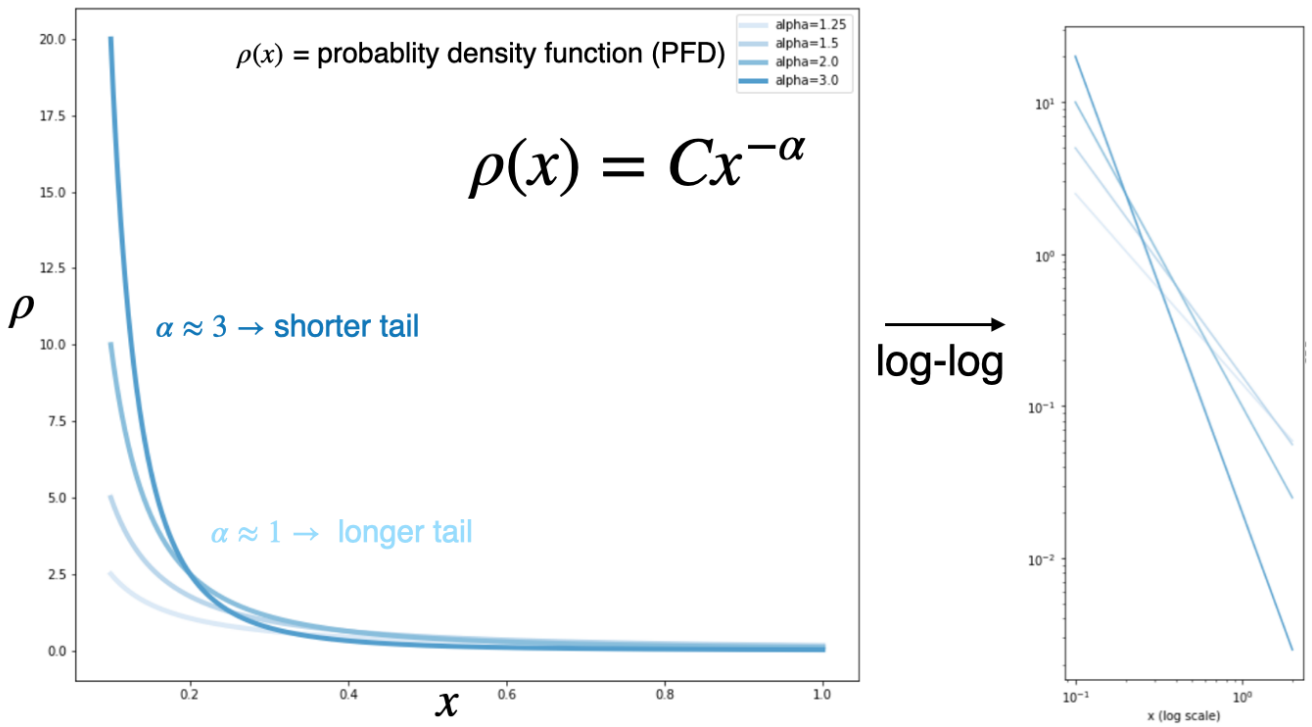


Figure 1: Power Law Distribution

3.4 Community Detection

Louvain Method:

- For networks with weighted co-occurrence edges, the Louvain algorithm is the gold standard.
- Maximizes modularity score Q across the network.
- Efficient and effective for large networks.

- Uses a two-phase iterative optimization process. In phase 1, each node is moved to the community that yields the highest gain in modularity. In phase 2, the algorithm aggregates nodes of the same community and builds a new network. These steps are repeated iteratively until modularity is maximized and cannot be increased further.

$$Q = \frac{1}{2m} \sum_{ij} \left[A_{ij} - \frac{k_i k_j}{2m} \right] \delta(c_i, c_j)$$

3.5 Vector Embeddings

TBD

4 Results

5 Discussion

6 Conclusion

7 References

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